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0.0.0.1 Section 1: Data Loading

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• Scripts:	
– load_gene_expression_data.R: Handles RPKM data loading.	
– load_gene_groups_data.R: Imports ortholog and paralog relationships.	
– load_gene_families_data.R: Processes MCL output for gene family clustering.	
• Input Data:	
• Expression Profiles	
– RNA-seq data across multiple tissues	
– Normalized expression values (TPM/RPKM)	
• Gene Groups	
– Conserved orthologs	
– In/out paralogs	
– Special in/out paralogs	
• Gene Families	
– MCL clustering output	
– Family-wise gene organization into Gene-Family Clusters	
– Formats and files: Expression Profiles (regular/log2), Gene Groups, Gene Families.	
– Link to the outputs from Chapter 3.	
– load Information about the Orthologs and Tandems identified within the eight Brassicaceae Genomes	
• the txt Files contain Information about Pairwise Sequence Similarities (Orthologs & Paralogs)	
– orthologs (genes in different species with a common ancestor), They typically retain the same function across species. genes from multiple species. These genes are found in the same cluster because they share a high degree of sequence similarity, which suggests that they are orthologs.	

1.0.1 1. load_gene_expression_data.R

Loads expression data for gene groups, typically in DNA FASTA format. Expression levels (e.g., RNA counts, RPKM) reflect gene activity: higher counts indicate active genes (ON), while lower counts indicate inactive

genes (OFF). Profiles are multi-dimensional vectors representing gene activity across tissues. Normalization methods (e.g., TPM, FPKM) adjust for technical variations, ensuring accurate comparisons between samples.

USAGE: Rscript exec/load_gene_expression_data.R <RPKM_counts_table.tsv>

1.0.2 2. load_gene_groups_data.R

Loads ortholog and paralog relationships across species, with files categorized by group type. Each file follows a specific header convention:

- **Conserved Orthologs:** One ortholog per species-gene pair; represents highly conserved, reliable relationships.
- **In Paralogs:** Species-specific paralogs from recent duplications with no conserved ortholog.
- **Special In Paralogs:** Species-specific paralogs with a conserved ortholog relationship, indicating potential conservation.
- **Out Paralogs:** Paralogs from older duplications, not tied to a conserved ortholog.
- **Special Out Paralogs:** Cross-species paralogs with a conserved ortholog relationship, suggesting retained functional ties.

1.0.2.1 con_orthologs.tsv

Family	Gene	Gene_species	Ortholog	Ortholog_species
OG00000000	FBpp0117097	dana	FBpp0172663	dmoj

1.0.2.2 in_paralogs.tsv

Family	Gene	Gene_species	Paralog	Paralog_species
OG00000000	FBpp0117097	dana	FBpp0172663	dmoj

1.0.2.3 special_in_paralogs.tsv

Family	Gene	Gene_species	Paralog	Paralog_species
OG00000000	FBpp0117097	dana	FBpp0172663	dmoj

1.0.2.4 out_paralogs.tsv

Family	Gene	Gene_species	Paralog	Paralog_species
OG00000000	FBpp0117097	dana	FBpp0172663	dmoj

1.0.2.5 special_out_paralogs.tsv

Family	Gene	Gene_species	Paralog	Paralog_species
OG00000000	FBpp0117097	dana	FBpp0172663	dmoj

1.0.2.6 3. load_gene_families_data.R Loads gene family clusters created by MCL. Each family clusters related genes based on sequence similarity, organizing them by shared evolutionary lineage.

- Clusters: In the context of gene families, a cluster refers to a group of genes that are related by sequence similarity and are assumed to have evolved from a common ancestor. Clusters represent groups of related genes, identified by their sequence similarity across different species
- Inputs: `mcl_output.txt` , `mcl_table.tsv` (from markov clustering tool mcl)
- Outputs: `families.lst` & `families.genes.df`, `families.df` -> `families.RData`

USAGE: Rscript exec/load_gene_families_data.R

1.0.2.7 <families_file>

Family	species1	species2	species3
family_1	gene1, gene2, gene3	gene4, gene5, gene6	gene7, gene8, gene9

USAGE: Rscript exec/load_gene_expression_data.R <RPKM_counts_table.tsv>

1.0.2.8 <RPKM_counts_table.tsv>

id	tissue	expression
gene1	tissue1	####
gene2	tissue1	####
gene1	tissue2	####
gene2	tissue2	####

- we need the mapping Information from the Fasta files to create rna.seq.exp.profiles, also to map the species name to the right gene we used the fasta file

1.0.2.9 <rna_seq_exp_profiles>

FBpp_ID	Parent_FBgn	Species	fat (tissue)	gut	muscle	whole_body
FBpp0291548	FBgn0085506	dmel	0.000000000	0.000000000	0.000000000	1.0000000
FBpp0289382	FBgn0259817	dmel	0.000000000	0.003827632	0.006253107	0.9899193
FBpp0312442	FBgn0085692	dmel	0.007196562	0.000000000	0.000000000	0.9928034
FBpp0077828	FBgn0002121	dmel	0.111506516	0.266536495	0.283942387	0.3380146
FBpp0111921	FBgn0031209	dmel	0.009854926	0.756561172	0.010681603	0.2229023

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2.1 RNA-Seq Data Normalization

2.1.1 Purpose

- Corrects technical biases in expression data
- Enables fair comparison across samples
- Accounts for sequencing depth variations

2.1.2 Common Methods

1. **RPKM** (Reads Per Kilobase Million)
 - Normalizes by gene length and library size
 - Formula: $(\text{reads} \times 10^6) / (\text{total_reads} \times \text{gene_length})$
 - Best for single-end sequencing
2. **FPKM** (Fragments Per Kilobase Million)
 - Similar to RPKM but for paired-end reads
 - Accounts for fragment rather than read counts
3. **TPM** (Transcripts Per Million)
 - More consistent across samples
 - Sums to same value (1 million) for each sample

2.1.3 Normalization Process

1. **Length Normalization**
 - Divides counts by gene length
 - Corrects for longer genes generating more reads
2. **Depth Normalization**
 - Scales to million mapped reads
 - Accounts for sequencing depth differences

2.1.4 Benefits

- Comparable expression values
- Reduced technical bias
- More accurate differential expression analysis
- **Normalization:** Adjusts raw expression data using RPKM methodology to ensure comparability.
- **Normalization of the Expression** (Transcriptome) data
- **Normalization Purpose:** Adjusts for technical variations (e.g., sequencing depth, RNA quality) to ensure fair comparison of gene expression levels across different samples.
- **Normalization Methods:** Common methods include TPM (Transcripts Per Million), FPKM (Fragments Per Kilobase of transcript per Million mapped reads), and RPKM (Reads Per Kilobase of transcript per Million mapped reads).
- **RPKM Overview:** RPKM is used in RNA sequencing to normalize gene expression by accounting for both gene length and total read depth, enabling comparison between different samples.
- **Gene Length Normalization:** RPKM divides read counts by the gene length (in kilobases), preventing longer genes from appearing more highly expressed simply due to their length.
- **Read Depth Adjustment:** To account for variations in total read counts between samples, RPKM scales read counts to a million mapped reads, ensuring fair comparison of gene expression levels across samples.